QUALITY INFORMATION DOCUMENT

For IBI Strong Wave Incidence index IBI_OMI_SEASTATE_swi

Issue: 1.2

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RECORD TABLE

When the quality of the products changes, the QuID is updated and a row is added to this table. The third column specifies which sections or sub-sections have been updated. The fourth column should mention the version of the product to which the change applies.

Issue	Date	§	Description of Change	Author	Validated By
1.0	2020/09/11	All	Initial version	Álvaro de Pascual Collar	
1.1	2021/02/23	references	Update of reference Pascual et al., 2020.	Álvaro de Pascual Collar	
1.2	2025/09/15	Sec. 1.c and 2	Review of the document, including updates to the Copernicus product identifiers	Álvaro de Pascual Collar	Copernicus Marine Product Management

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1) INTRODUCTION

a) Product Name

IBI_OMI_SEASTATE_swi

b) Indicator Evaluation Method

The Strong Wave Incidence index (SWI) is proposed to quantify the variability of strong wave conditions in the Iberia–Biscay–Ireland (IBI) regional seas. The anomaly in the exceedance of a significant wave height (SWH) threshold is used to characterize wave behavior. A sensitivity analysis was performed to evaluate the impact of using different thresholds (75th, 80th, 85th, 90th, and 95th percentiles). Based on this analysis, the 90th percentile was selected as the most representative, aligning with current state-of-the-art practices.

The robustness of the proposed indicator has been assessed through:

1. Comparison with the North Atlantic Oscillation (NAO) index,
2. Verification of anomalous events detected in recent decades against documented events in the literature, and
3. Analysis of the atmospheric patterns associated with the occurrence of intense wave events (Pascual et al., 2020).

c) Copernicus Marine products used as input for the computation

Product name (and Type)	Documentation
IBI_MULTIYEAR_WAV_005_006 Numeriacl Models (reanalysis)	QUID: https://documentation.marine.copernicus.eu/QUID/CMEMS-IBI-QUID-005-006.pdf PUM: https://documentation.marine.copernicus.eu/PUM/CMEMS-IBI-PUM-005-006.pdf
IBI_ANALYSISFORECAST_WAV_005_005 Numerical Models (analysis and forecast)	QUID: https://documentation.marine.copernicus.eu/QUID/CMEMS-IBI-QUID-005-005.pdf PUM: https://documentation.marine.copernicus.eu/PUM/CMEMS-IBI-PUM-005-005.pdf

2) SCIENTIFIC VALIDATION

The validation of the IBI_OMI_SEASTATE_swi indicators is based on the quality assessment of the underlying model data:

- CMEMS product IBI_MULTIYEAR_WAV_005_006
  - QUID: https://documentation.marine.copernicus.eu/QUID/CMEMS-IBI-QUID-005-006.pdf
- CMEMS product IBI_ANALYSISFORECAST_WAV_005_005
  - QUID: https://documentation.marine.copernicus.eu/QUID/CMEMS-IBI-QUID-005-005.pdf

Additionally, product consistency has been evaluated in Pascual et al. (2020), with key findings summarized below. The correlation between the Strong Wave Incidence index and the NAO index (**Figure 1**) yielded results consistent with previous studies. Furthermore, the SWI successfully identified historical wave events reported in the literature, such as the extreme event in early 2014 that affected the European coast from Ireland to Cape St. Vincent. Finally, the SWI has demonstrated its ability to describe atmospheric patterns that drive intense wave events across the four defined subregions.

These results support the physical consistency of the SWI and suggest that the index not only reflects variability in significant wave height but also captures the underlying physical processes driving wave behavior in the IBI region.

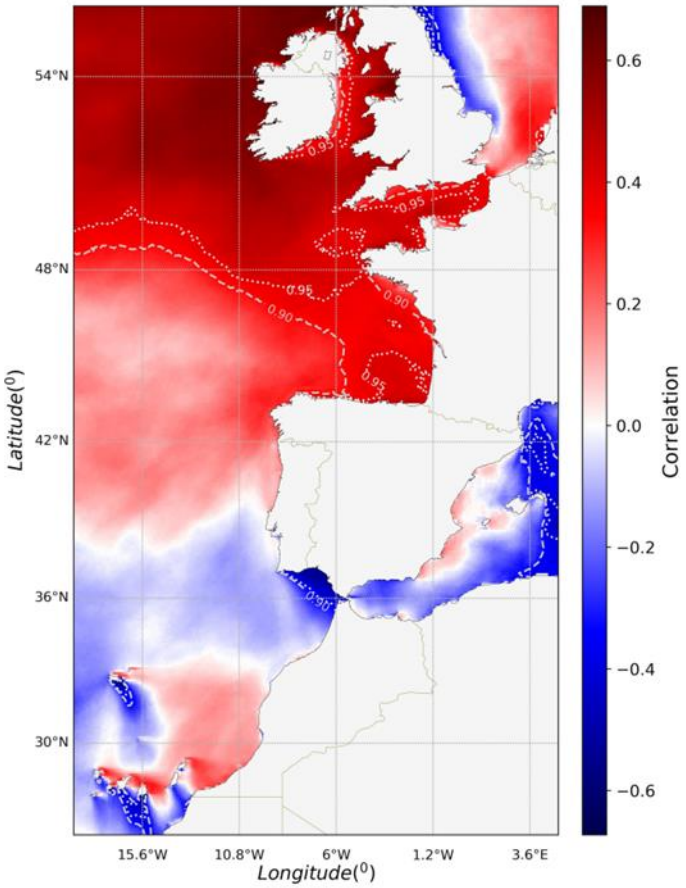

Figure 1 : *Temporal correlation coefficient between annual Strong Wave Incidence index (computed from IBI-WAV-MYP product) and North Atlantic Oscillation index (NOAA National Weather Service, 2005). Dashed and dotted contour lines represent the minimum statistical significance of the correlation values (levels 90% and 95% respectively).*

Further details on the consistency evaluation and a full description of the indicator can be found in de Pascual et al. (2020).

3) REFERENCES

de Pascual A., Levier B., Aznar R., Toledano C., García-Valdecasas JM., García M., Sotillo M., Aouf L., Álvarez E. (2020) Monitoring of wave sea state in the Iberia-Biscay-Ireland regional seas. In von Scuckmann et al. (2020) Copernicus Marine Service Ocean State Report, Issue 4, Journal of Operational Oceanography, 13:sup1, S1-S172, DOI: 10.1080/1755876X.2020.1785097